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Cognitive functioning following traumatic brain injury in older adults: associations with social participation and health-related quality of life

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ABSTRACT

Objectives: To describe objective and subjective cognitive functioning older adults who sustained TBI at age 65 or over, and to determine whether cognitive functioning is associated with health-related quality of life (HRQoL) and social participation.

Method: The sample consisted of 40 individuals with TBI (mean age = 73 years; 65% mild, 35% moderate/severe TBI). On average 15 months post-injury, they completed measures of objective and subjective cognitive functioning (Telephone Interview for Cognitive Status-Modified, Alphaflex, Medical Outcomes Study Cognitive Functioning Scale), HRQoL (SF-12), and social participation (Participation Assessment with Recombined Tools – Objective).

Results: Mean score for objective cognitive functioning was lower than normative values, while mean scores for executive functioning and subjective cognitive functioning were comparable to normative values. There was no relationship between objective and subjective measures. Subjective cognitive functioning and (to a lesser extent) global objective cognitive functioning were significantly associated with mental HRQoL but not with physical HRQoL or social participation.

Conclusion: These results underscore the importance of considering both subjective perception and objective performance when assessing and intervening on cognition to promote better mental HRQoL in older adults with TBI.

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Introduction

Traumatic brain injury (TBI) is prevalent in the elderly population (1). It is also expected to increase over the next few years, in large part due to the aging of the population as older adults are more vulnerable to falls that can cause TBI or spinal cord injury (2). Indeed, in Canada, it is estimated that the number of TBI-related hospitalizations in adults over 65 years will have increased by 28% between 2014 and 2024 (3). Understanding the impact of TBI in older adults is thus becoming increasingly important.

Regardless of age, TBI may lead to significant physical, behavioral, affective, and cognitive consequences (4). Impairments in memory, attention, and executive functioning have been well documented, mainly as a result of damage to the anterior brain regions (5). The magnitude and persistence of cognitive impairments also generally increase as a function of TBI severity (6). Compared to other health-related conditions such as chronic pain or spinal cord injury, individuals with TBI report worse cognitive impairments (7,8). Moreover, older adults show worse cognitive outcomes following TBI than younger adults (9,10).

Few studies have focused on cognitive functioning after TBI specifically in older adults, especially in the first 1–2 years (2). In a systematic review of mild TBI in older people only two of the included studies investigated cognition and results showed that cognitive deficits may still be present 1 year post-injury (11). In adults, TBI severity affects self-reported assessment of cognitive functioning. Reduced self-awareness is frequently observed following moderate-to-severe TBI (12,13) and may explain the frequent discrepancy between cognitive complaints and results of objective testing. Conversely, adults with mild TBI often perceive their impairments as more severe than what is observed objectively (14,15). This has been linked to emotional aftereffects, concomitant physical injuries, intensity of daily demands, lack of social network availability (14), and motivational factors such as financial compensation (15).

Cognitive complaints are common in older adults without TBI. While subjective cognitive decline has been shown to increase the risk of mild and major neurocognitive disorder (16), worse subjective than objective cognitive performance may also be partly related to a decline in metacognitive capacities associated with normal aging (17). For older adults who have sustained TBI, it could be hypothesized that the overlap between

reduced metacognitive abilities resulting from normal aging and the differential effect of TBI severity on subjective cognitive assessment would lead to a dissociation between subjective and objective assessment of their cognition. To our knowledge, this relationship has not yet been explored in this population.

Cognitive sequelae following TBI are known to be associated with long-term disability and social participation (18), interfering with leisure activities, social relations, and activities of daily living (19). It has been proposed that objective cognitive functioning and, more importantly, subjective perception of cognitive functioning, may influence these outcomes (20). However, research has yet to determine whether objective and subjective cognitive functioning might be related to quality of life and social participation in older adults with TBI.

The occurrence of TBI is associated with a significant decrease in health-related quality of life (HRQoL), defined as the perception of how a condition affects physical, mental, and social life aspects (21). Although several factors can contribute to HRQoL after TBI, such as diminished physical health or elevated fatigue, cognitive impairments could also play a significant role. A positive association between cognitive functioning and HRQoL has been shown in individuals with other conditions including multiple sclerosis, nephrological disorders, and pathological aging (22–24).

In addition to its impact on HRQoL, TBI can also affect social participation, i.e., the realization of life habits according to the interaction between personal and environmental factors (25). Social participation tends to decrease with age regardless of health status (26), and older age has been described as a predictor of worse social participation after TBI (27). Moreover, some studies in adults with TBI have reported a positive association between social participation after TBI and objective and subjective cognitive functioning (28,29). Nonetheless, this relation has not yet been explored specifically in older adults after TBI.

Despite emerging interest in the literature, still little is known regarding the relationships between objective and subjective cognitive functioning, social participation, and HRQoL in older adults with TBI. Thus, the specific objectives of the present study are (1) to describe subjective and objective cognitive functioning in older adults having sustained a TBI after 65 years of age and (2) to determine whether objective and subjective cognitive functioning are related to HRQoL and social participation. It is hypothesized that both objective and subjective cognitive functioning will be positively related to social participation and HRQoL but more strongly so for subjective cognitive functioning.

Methods

This cross-sectional study is part of a larger single-cohort longitudinal research project on the psychological consequences of TBI in older adults and was approved by the Ethics Research Board of the CHU de Québec – Université Laval (#116069).

Participants

The sample included individuals who were hospitalized in a Level I trauma center (CHU de Québec) after a mild to severe TBI between May 2017 and February 2019. Participants had to be aged 65 years and over at the time of TBI, and data collection took place in the following 2 years (see Procedure section). Injury severity was determined by a physician at the trauma center based on provincial guidelines (criteria include duration of loss of consciousness and post-traumatic amnesia, presence of neurological signs, brain imaging results, and initial score on the Glasgow Coma Scale GCS). Those with concomitant spinal cord injury or stroke, with a past history of moderate or severe TBI or other neurological disorder, who were not fluent in French, had significant hearing difficulties, or were unable to provide informed consent were excluded.

Procedure

At the trauma center (CHU de Québec), a nurse presented an information pamphlet to eligible participants and obtained their verbal consent to be contacted approximately 3 months post-TBI. For the main research project, participants completed interviews and questionnaires 4, 8, and 12 months post-TBI. The present study only used data from the social participation and quality of life measures of the 12-month questionnaire. Additionally, a trained graduate student in clinical neuropsychology (LC) contacted participants by phone 12 months post-TBI to obtain their verbal consent to participate in a brief add-on to the main research project involving a telephone evaluation of objective and subjective cognitive functioning (20 min). Participants were asked to be alone in a non-distracting environment and not to use any external aid (e.g., paper, pencil, calendar, smartphone). Participants received a \$20 gift card for the cognitive assessment. All data were collected before the implementation of sanitary measures related to the COVID-19 pandemic in Canada (March 2020).

Measures

Sociodemographic and clinical information

Sociodemographic information such as age, sex, and level of education was collected during the initial recruitment contact. Mechanism of injury and TBI severity as determined by the medical team at the trauma center were extracted from the medical files.

Subjective cognitive functioning

The Medical Outcomes Study Cognitive Functioning Scale (MOS-Cog) is a recommended tool for research use with good internal consistency ($\alpha = 0.87$) (30). It includes six items assessing the frequency of subjective cognitive difficulties (i.e., reasoning, concentration, confusion, forgetfulness, maintaining attention, reacting slowly) over the previous 4 weeks, scored on a five-point Likert scale (i.e., 1 = all of the time; 5 = never). Mean raw scores are transformed into a 20–100 scale, with higher scores indicating better subjective cognitive functioning. Comparisons were performed with the normative

data of 3,053 individuals aged 18–98 years ($M = 82.40$; $SD = 16.50$) (30). Significant cognitive difficulties were defined as a score lower than one standard deviation (SD) below the normative mean of the whole validation sample (i.e., <65.90) as age-stratified norms are not available.

Objective cognitive functioning

The Telephone Interview for Cognitive Status-Modified (TICS-m) is a brief standardized screening test for global objective cognitive functioning administered over the phone (<10 min), mostly used with the elderly population (31,32). The validated French version (F-TICS-m) used for the present study demonstrates higher sensitivity and specificity (33) than the Mini Mental State Examination (34). The F-TICS-m includes 17 items evaluating temporal (5 points) and spatial (3 points) orientation, immediate (10 points), and delayed (10 points) verbal episodic memory, attention and calculation (5 points), semantic memory (6 points), language (3 points), and praxis (1 point). As suggested by the authors of the TICS-m, the word list was presented only once before immediate retrieval (31,32), which differs from Vercambre and colleagues' (33) administration (three repetitions of the word list). Slight modifications were made to the content of the F-TICS-m to make it more culturally appropriate for French Canadian individuals (35): in the word list "tenaille" (pliers; seldom used in Canada) was replaced by "marteau" (hammer), and for semantic memory "current and previous Presidents of the French Republic" were replaced by "current and previous Prime Ministers of Canada." Item scores and total score (ranging from 0 to 43) were transformed into a 0–100 scale, higher scores indicating better objective cognitive functioning. Significant cognitive difficulties were qualified as a total score lower than one SD below the mean of the "cognitively normal" group of the validation sample of the F-TICS-m ($M = 78.84$; $SD = 6.51$; i.e., <72.33) (33). This cutoff also presents the best trade-off between sensitivity and specificity (33). Since the number of repetitions of the word list in the present study differs from the validation study (33), and thus lower results are expected, more stringent cutoffs of -1.5 (<69.08) and -2 (<65.82) SDs below the mean were also used as comparators.

Executive functioning is not assessed by the F-TICS-m but is known to be impaired after TBI (36). The Alphaflex (37) was therefore used to assess cognitive flexibility, working memory, and inhibition. It was specifically developed to be administered over the phone with the elderly population and has shown low to moderate correlations with the Trail Making Test, a widely used test evaluating the same components of executive functioning (37). In condition A of the Alphaflex, participants are asked to recite the complete sequence of the alphabet. In condition B, participants are asked to recite the alphabet again but this time by alternating between the oral production of a letter and the subvocalization of the next one. The completion time and errors differences between the first and second condition ($B - A$) were used. Performances below the 10th and 5th percentile (considered impaired) from the original study (37) were considered in the present research.

Health-related quality of life

The Medical Outcomes Study 12-Item Short Form Health Survey (SF-12) is a self-report questionnaire of HRQoL (38). Two component scores are derived from eight scales: the physical health component score (PCS) from the physical functioning, role limitations due to physical health problems, bodily pain, and general health subscales; and the mental health component score (MCS) from the vitality, social functioning, role limitations due to emotional health problems, and general mental health subscales. The raw component scores are then transformed into a 0–100 scale, with higher scores indicating greater HRQoL. Various studies have reported adequate validity and reliability of the SF-12 (39).

Social participation

The Participation Assessment with Recombined Tools-Objective (PART-O) (40) is a 17-item self-report questionnaire specifically developed for the TBI population. It is used to measure the extent to which an individual objectively engages in a range of activities covering three domains of social participation: productivity, social relations, and community involvement. The total score is derived from the average domain scores and ranges between 0 and 5, higher scores indicating greater social participation. The PART-O presents good internal consistency as well as good convergent validity with the Mayo-Portland Adaptability Inventory ($r = 0.69$) (40).

Statistical analyses

All statistical analyses were performed using the SPSS software, versions 25 and 27 (IBM Corp, 2017, 2020). Alpha level was set at bilateral 0.05. As data were not normally distributed non-parametric exact Mann-Whitney U tests and Spearman correlations were preferred over parametric t -tests and Pearson correlations. For objective 1, descriptive statistics were used to document objective and subjective cognitive functioning, and scores were compared to available comparative samples. Exact probability Mann-Whitney U tests were then performed to investigate differences between subgroups of interest, i.e., age (<75 years vs ≥ 75 years), sex (men vs women), level of education (≤ 12 years vs > 12 years), mechanism of injury (fall vs other mechanisms), and TBI severity (mild vs moderate/severe). Confidence intervals (95% CI) were estimated using a bias-corrected bootstrap technique with 2000 iterations. Effect sizes ($r = Z/\sqrt{N}$) were computed and interpreted using Cohen's criteria (41). To determine how objective and subjective cognitive functioning were related, nonparametric Spearman correlations were performed and a scatterplot with results from the F-TICS-m and MOS-Cog was examined. For objective 2, Spearman correlations were computed to explore the associations between cognitive functioning (MOS-Cog, F-TICS-m, Alphaflex response time and errors), and HRQoL and social participation measures (SF-12 MCS, PCS; PART-O). Multiple hierarchical linear regressions were then conducted with variables that were correlated at $p < 0.10$, with order of entry of predictors based on theoretical relevance and results of correlations.

Results

Sample description

The final sample of the present study consisted of 40 participants (see Figure 1). After initial contact at the trauma center, 135 individuals agreed to be contacted to obtain more information regarding the larger study (79% of eligible individuals), and, of those, 82 (61%) were enrolled. During the first year post-TBI, 4 participants were excluded and 12 were lost. Among the 66 remaining participants one-year post-TBI, 52 (79%) were reached by phone and 40 (61%) agreed to take part in the current study (i.e., cognitive assessment). Participants completed the phone interview with the subjective and objective cognitive measures on average 15.38 months after TBI ($SD = 2.81$; range = 12–21). The SF-12 and PART-O were completed on average within 3 months of this cognitive assessment except for two participants who did not complete the questionnaires. Sociodemographic and injury characteristics of participants are presented in Table 1. Individuals who declined to participate or were unreachable ($n = 26$) had a significantly lower level of education ($M = 10.15$, $SD = 3.96$ years) compared to participants of the current study ($n = 40$, $M = 13.40$, $SD = 4.85$), $U = 244$, $p = 0.01$, $r = -0.32$. The groups did not differ on age, sex, mechanism of injury, or TBI severity ($ps > 0.05$).

Subjective cognitive functioning

Results on cognitive measures are presented in Table 2. MOS-Cog scores ranged between 40 and 100, with a mean of 74.50, which is -0.48 SD below the normative mean (30). Moreover, 30% of the sample had scores below the cutoff of 65.90 (normative mean -1 SD). Mean scores for individual items ranged from 69.00 (forgetfulness) to 76.60 (confusion and reasoning). Participants with 12 years or less of education ($M = 69.05$, $SD = 16.20$, *bootstrapped* 95% $CI = 62.40$ – 75.70 , $n = 21$) had significantly lower self-reported cognitive functioning on the MOS-Cog compared to those with more than 12 years of education ($M = 80.52$, $SD = 12.68$ *bootstrapped* 95% $CI = 74.99$ – 86.16 , $n = 19$), $U = 121$, $p = 0.03$, $r = -0.34$ (moderate effect size). No differences on subjective cognitive functioning were found for sex, age, TBI severity, or mechanism of injury ($ps > 0.44$; $r < .13$).

Objective cognitive functioning

F-TICS-m. Mean F-TICS-m score (61.09; $SD = 12.28$; range: 18.60–81.40) was -2.73 SD below the mean of “cognitively normal” participants, -1.29 SD below the mean of participants with mild cognitive impairment (MCI) and 0.57 SD above that of participants with dementia included in the French validation sample (33). A total of 83% of our sample scored below one SD below the mean of “cognitively normal” participants of the

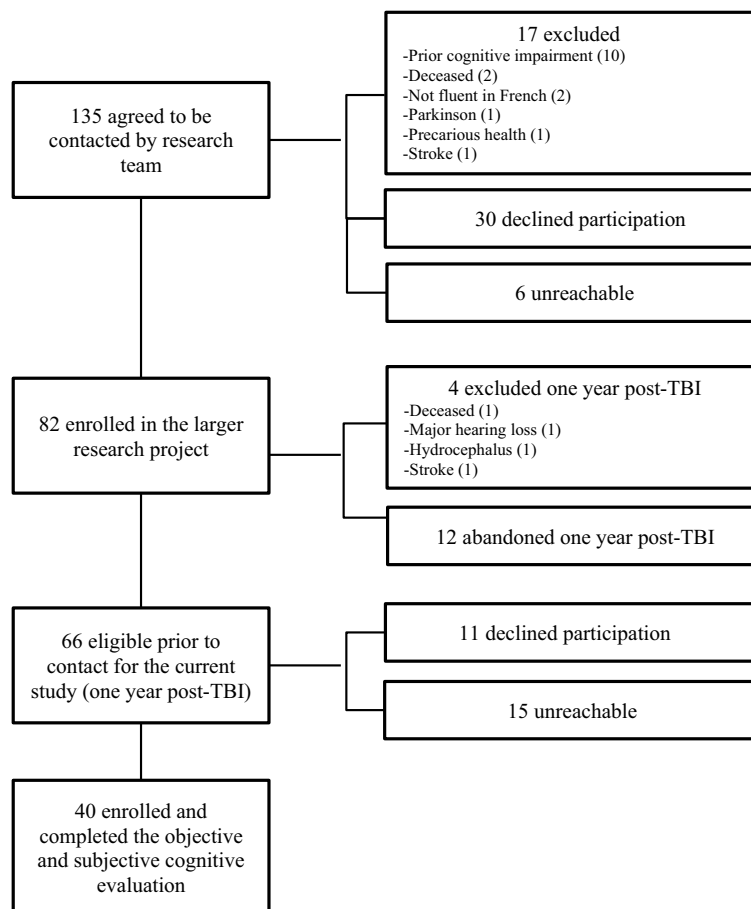


Figure 1. Flow chart of participants recruitment.

Table 1. Sociodemographic and injury-related characteristics.

Characteristics	Value
Age, M $\pm$ SD(range)	73 $\pm$ 4.78 (66–84)
<75 years old, n (%)	26 (65.0)
$\geq$ 75 years old, n (%)	14 (35.0)
Education, M $\pm$ SD(range)	13.40 $\pm$ 4.85 (6–25)
$\leq$ 12 years (%)	21 (52.5)
>12 years (%)	19 (47.5)
Female – n (%)	10 (25.0)
TBI severity – n (%)	
Mild	26 (65.0)
Moderate	13 (32.5)
Severe	1 (2.5)
Mechanism of injury – n (%)	
Fall	21 (52.5)
Motor vehicle/Traffic accident	10 (25.0)
Hit by an object	5 (12.5)
Sport/Recreational activity	3 (7.5)
Criminal act assault	1 (2.5)

Abbreviations: M=mean; SD = standard deviation; TBI = traumatic brain injury.

validation study (<72.33), and this proportion was 73% and 70% for more stringent cutoffs of -1.5 SD (<69.08) and -2 SD (<65.82) (33). To facilitate comparison, all individual item scores were transformed into a 0–100 scale. Mean scores of individual items were, from lowest to highest: delayed recall (28/100), immediate recall (43/100), language (59/100), semantic memory (79/100), spelling backwards (80/100), calculation (81/100), tapping 5 times (90/100), temporal orientation (93/100), and spatial orientation (94/100). No significant between-group differences were found on the F-TICS-m total score for sex, age, level of education, TBI severity, or mechanism of injury (p s > 0.06; r s < |.30|).

Alphaflex

The mean response time difference score between conditions A and B (9.98 s; SD = 11.84; range: -15 – 64) was -0.41 SD below the normative mean of 13.50 seconds (SD = 8.50) (37),

higher differences indicating poorer performance. The mean errors difference score between condition A and B (1.90; SD = 2.58; range: 0–9) was 0.53 SD above the normative mean (37), higher differences indicating poorer performance. The percentages of participants scoring below the 10th percentile were 10% (n = 4) for the time difference and 17.5% (n = 7) for the errors difference, and the percentages scoring below the 5th percentile were 5% (n = 2) and 15% (n = 6), respectively. No significant differences were found on the Alphaflex response time or error difference scores for sex, age, level of education, TBI severity, or mechanism of injury (p s > 0.12; r s < |.25|).

Relationship between objective and subjective cognitive performance

The MOS-Cog total score was not significantly correlated with the F-TICS-m total score (ρ = 0.05, p = 0.76), or Alphaflex response time (ρ = -0.02 , p = 0.91), or error (ρ = -0.23 , p = 0.16) difference score. The scatterplot of the relationship between objective (F-TICS-m) and subjective (MOS-Cog) cognitive functioning is presented in Figure 2.

Associations between cognition, HRQoL, and social participation

Spearman correlations computed between cognitive functioning, HRQoL, and social participation measures are presented in Table 3. The MOS-Cog total score was significantly correlated with the MCS (ρ = 0.59, suggesting a large effect size, p < 0.001), better subjective cognitive functioning being associated with better mental HRQoL. The only other correlation with a p value < 0.10 was found between the F-TICS-m and MCS (ρ = -0.29 , suggesting a moderate to large effect size, p = 0.08), with poorer objective cognitive functioning

Table 2. Exact probability Mann–Whitney U tests for subgroup comparisons on the MOS-Cog, F-TICS-m and Alphaflex and mean individual item's score of the F-TICS-m.

Variables	M $\pm$ (SD)	U	p	r
MOS-Cog	74.50 (15.57)			
Sex (women vs men)	74.32 $\pm$ (16.18) vs 74.55 $\pm$ (15.64)	146.50	.92	-.02
Age (65–74 vs 75–84 years)	73.59 $\pm$ (15.52) vs 76.18 $\pm$ (16.11)	164.50	.63	-.08
Education ($\leq$ 12 vs >12 years)	69.05 $\pm$ (16.20) vs 80.52 $\pm$ (12.68)	121.00	.03*	-.34
Mechanism (fall vs others)	76.51 $\pm$ (14.85) vs 72.27 $\pm$ (16.44)	171.00	.45	-.12
TBI severity (mild vs others)	73.71 $\pm$ (14.83) vs 75.96 $\pm$ (17.35)	164.00	.62	-.08
F-TICS-m total score	61.09 (12.28)			
Sex (women vs men)	66.04 $\pm$ (12.02) vs 59.47 $\pm$ (12.10)	105.50	.17	-.22
Age (65–74 vs 75–84 years)	62.60 $\pm$ (10.50) vs 58.30 $\pm$ (15.06)	150.00	.37	-.14
Education ($\leq$ 12 vs >12 years)	58.02 $\pm$ (12.51) vs 64.51 $\pm$ (11.36)	133.00	.07	-.29
Mechanism (fall vs others)	59.26 $\pm$ (14.99) vs 63.16 $\pm$ (8.25)	167.00	.38	-.14
TBI severity (mild vs others)	63.76 $\pm$ (9.32) vs 56.14 $\pm$ (15.63)	133.00	.17	-.22
Alphaflex (time)	9.98 (11.84)			
Sex (women vs men)	13.00 $\pm$ (19.40) vs 8.97 $\pm$ (8.21)	130.00	.54	-.10
Age (65–74 vs 75–84 years)	9.88 $\pm$ (13.84) vs 10.14 $\pm$ (7.22)	149.50	.36	-.15
Education ($\leq$ 12 vs >12 years)	9.71 $\pm$ (14.92) vs 10.26 $\pm$ (7.48)	174.50	.51	-.11
Mechanism (fall vs others)	12.10 $\pm$ (14.12) vs 7.63 $\pm$ (8.43)	183.50	.67	-.07
TBI severity (mild vs others)	9.77 $\pm$ (14.09) vs 10.36 $\pm$ (6.18)	150.50	.38	-.14
Alphaflex (errors)	1.90 (2.58)			
Sex (women vs men)	1.06 $\pm$ (0.34) vs 2.81 $\pm$ (0.51)	103.00	.13	-.24
Age (65–74 vs 75–84 years)	1.61 $\pm$ (2.17) vs 2.43 $\pm$ (3.23)	159.50	.51	-.11
Education ($\leq$ 12 vs >12 years)	2.19 $\pm$ (2.66) vs 1.57 $\pm$ (2.52)	162.00	.29	-.17
Mechanism (fall vs others)	1.80 $\pm$ (2.69) vs 2.00 $\pm$ (2.52)	176.50	.52	-.10
TBI severity (mild vs others)	1.88 $\pm$ (2.57) vs 1.92 $\pm$ (2.69)	179.00	.94	-.01

Abbreviations: F-TICS-m = French Telephone Interview for Cognitive Status modified; MOS-Cog = Medical Outcomes Study Cognitive Functioning Scale; TBI = traumatic brain injury; M = mean; SD = standard deviation.

* p < 0.05.

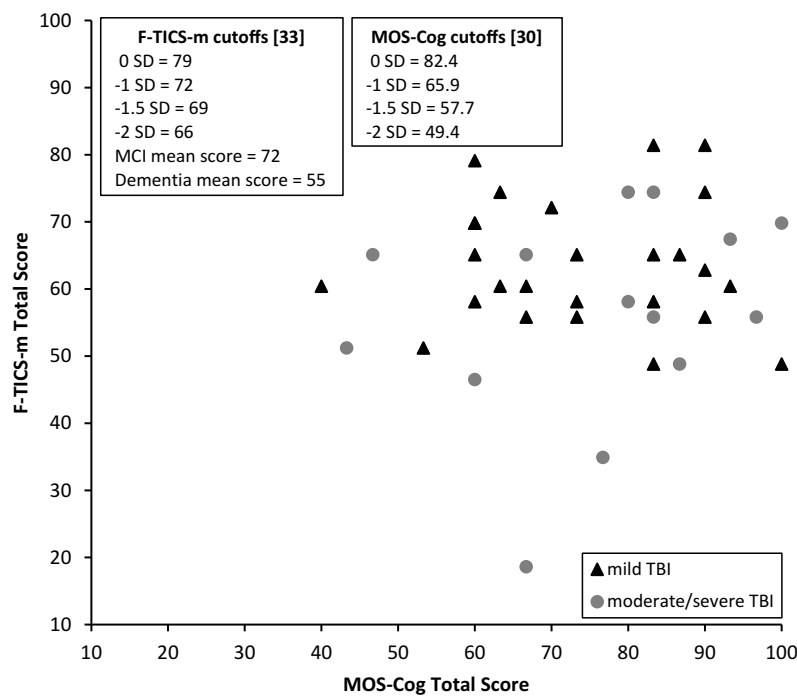


Figure 2. F-TICS-m and MOS-Cog total score scatterplot as a function of TBI severity. Abbreviations: F-TICS-m = French Telephone Interview for Cognitive Status modified; MCI=mild cognitive impairment; MOS-Cog = Medical Outcomes Study Cognitive Functioning Scale; SD = standard deviation .

Table 3. Spearman correlation matrix between HRQoL (SF-12), social participation (PART-O) and cognitive functioning (MOS-Cog, F-TICS-m and Alphaflex).

Variable	PCS	MCS	PART-O
MOS-Cog	$\rho = 0.26, p = 0.11$	$\rho = 0.59, p = 0.001^*$	$\rho = 0.23, p = 0.16$
F-TICS-m	$\rho = 0.14, p = 0.41$	$\rho = -0.29, p = 0.08$	$\rho = 0.02, p = 0.89$
Alphaflex (time score)	$\rho = 0.21, p = 0.20$	$\rho = -0.05, p = 0.78$	$\rho = 0.07, p = 0.68$
Alphaflex (error score)	$\rho = -0.04, p = 0.80$	$\rho = 0.14, p = 0.42$	$\rho = 0.17, p = 0.31$

Abbreviations: F-TICS-m = French Telephone Interview for Cognitive Status modified; HRQoL = health-related quality of life; MCS = SF-12 Mental Component Score; MOS-Cog = Medical Outcomes Study Cognitive Functioning Scale; PART-O = Participation Assessment with Recombined Tools-Objective; PCS = SF-12 Physical Component Score.

* $p < 0.05$.

associated with better mental HRQoL. As no correlations with $p < 0.10$ were found between cognitive functioning and either physical HRQoL (PCS) or social participation (PART-O), only one multiple hierarchical linear regression was performed to further examine the relationship between mental HRQoL and cognitive functioning (see Table 4). Assumptions of multiple regression were tested and parameters were within normal range for multicollinearity, independence of residuals, homoscedasticity, linearity, and normality of residue distribution (42). Examination of Mahalanobis distance revealed one multivariate outlier ($p = 0.0009$), but it was retained in the analysis as it did not influence the regression model. The MOS-Cog total score was entered in the first step and the F-TICS-m total score in the second step. A significant regression equation was found ($F(2, 35) = 17.38, p < 0.001, R^2 = 0.50, R^2_{adj} = 0.47$), with nearly

half of the variance of the MCS explained by the MOS-Cog and F-TICS-m scores. The unique contribution was 41% for subjective functioning (MOS-Cog, $\beta = 0.66$; positively related to mental HRQoL) and 9% for objective functioning (F-TICS-m; $\beta = -0.30$; negatively related to mental HRQoL), ($ps < 0.05$). A similar regression equation was obtained if the multivariate outlier was excluded, ($F(2, 34) = 16.70, p < 0.001, R^2 = 0.50, R^2_{adj} = 0.47$).

Discussion

The purpose of this study was to describe objective and subjective cognitive functioning of individuals aged 65 and older who had sustained a TBI on average 15 months (12–21) before, and to explore the relationship between cognitive functioning,

Table 4. Results of the multiple hierarchical linear regression for HRQoL (SF-12 MCS) ($n = 38$).

Predictors	B	β	t	p	sr^2	R^2	R^2_{adj}
Step 1							
MOS-Cog	0.34	0.64	4.96	<0.001		0.41	0.39
Step 2							
MOS-Cog	0.35	0.66	5.47	<0.001	0.41	0.50	0.47
F-TICS-m	-0.48	-0.30	-2.53	0.02	0.09		

Abbreviations: F-TICS-m = French Telephone Interview for Cognitive Status modified; HRQoL = health-related quality of life; MCS = SF-12 Mental Component Score; MOS-Cog = Medical Outcomes Study Cognitive Functioning Scale.

HRQoL, and social participation. Performance on a telephone cognitive screening test was significantly lower than that of available normative data from healthy older adults without TBI (33). Subjective cognitive functioning and performance on an executive functioning test were however within normal limits. Objective and subjective cognitive functioning measures were not significantly correlated. In addition, better subjective cognitive functioning, and to a lesser extent poorer objective cognitive functioning, were associated with better mental HRQoL. Objective or subjective cognitive functioning was not associated with physical HRQoL or social participation.

Subjective cognitive functioning was found to be similar to that of a normative group who did not sustain a TBI ($z = -0.48$) (30). This result is in agreement with the findings of Sigurdardottir and colleagues (43) who suggested that adults with TBI had no or mild cognitive complaints 12 months post-injury. Contrary to previous studies in adults (7,14,15), subjective cognitive functioning did not differ as a function of TBI severity (effect size below the small effect threshold). This highlights the fact that subjective cognitive functioning assessment might be independent of TBI severity in older adults. This may be due in part to the fact that older adults, regardless of TBI severity, are less likely to be exposed to situations of daily living that may reveal potential cognitive deficits compared to younger adults. However, a dichotomous variable was used (mild vs moderate/severe TBI) as other markers of severity such as duration of posttraumatic amnesia were not available. Furthermore, subjective cognitive functioning may be more sensitive to change from pre- to post-TBI than objective functioning although this assumption cannot be verified in our study because of the lack of pre-injury cognitive measures. Our results also showed that better subjective cognitive functioning is associated with higher education (moderate effect size), which has been reported in older adults without TBI (44). This association may indicate higher levels of cognitive reserve through higher educational-level attainment (45). By maintaining a normal level of cognitive functioning for a longer time individuals with more education may be able to cope better with the impact of brain damage resulting from TBI (46).

Global objective cognitive functioning was found to be between individuals with MCI and those with dementia from the F-TICS-m validation study (33) but is probably closer to the MCI group since the administration procedure of the memory items differed and may have resulted in poorer performance. However, Cook et al. (47) observed a mean total TICS-m score in individuals with MCI ($M = 62.40$, $SD = 7.00$) that was very similar to our sample ($M = 61.09$, $SD = 12.28$). This supports the hypothesis that cerebral changes associated with normal aging combined with damage resulting from TBI may lead to increased cognitive decline (18). Our results also indicate that the majority of participants (70% to 83% depending on the cutoff) had significant objective cognitive difficulties on a screening test. This rate is much higher than in a younger sample with TBI (average of 51 years) (48) in which 33% were cognitively impaired on the TICS-m but similar to another study in which 66% was considered cognitively impaired 1 year after mild to severe TBI (49). Our finding may suggest

TBI consequences on objective cognitive functioning, but it cannot be ruled out that some participants may have been on the spectrum of pathological aging in the form of a subjective cognitive decline or even MCI.

Executive functioning in our sample was found to be similar to the normative group who did not sustain a TBI, suggesting no significant executive impairments on the Alphaflex. This finding contrasts with other studies showing that performance on the Trail Making Test (TMT), a widely used measure of executive functioning requiring psychomotor speed and visual scanning (contrary to the Alphaflex), was significantly lower in people with TBI compared to controls (50,51). Visual and visuomotor impairments following TBI (52) may partially explain the discrepancy between TMT and Alphaflex results. However, one limitation of the Alphaflex is that a large majority of healthy older adults perform well on this test (37). In addition to this potential lack of sensitivity, the divergent results between the F-TICS-m and the Alphaflex may be explained by the fact that 47 points out of 100 on the F-TICS-m are related to memory.

Our findings also suggest that subjective cognitive functioning was not significantly correlated with objective cognitive functioning (effect sizes below the small effect threshold for F-TICS-m and Alphaflex response time, between small and moderate threshold for Alphaflex error score). This is consistent with several studies in adults with TBI (53,54), thus providing additional support that subjective report does not necessarily reflect underlying performance. In the general elderly population, it has been suggested that reduced meta-cognitive capacities could account for the lack of association between objective and subjective measures (17). Also, older individuals may more often rely on accumulated knowledge and habits rather than processing novel information unlike what is seen in cognitive assessment where several novel tasks are presented. The lack of exposure to deficit-revealing situations may also explain the apparent discrepancy between objective and subjective cognitive functioning (55). Moreover, it is possible that the MOS-Cog and the F-TICS-m measure different aspects of cognition, thus explaining the lack of association. For example, approximately half of F-TICS-m score is determined by memory performance, while memory is only assessed in one of the six MOS-Cog items.

This study showed that subjective cognitive functioning is significantly associated with mental HRQoL (large effect size in univariate analysis, 41% explained variance of mental health HRQoL in multivariate analysis). Similarly, Pusswald et al. (24) found that individuals without TBI but with subjective cognitive decline, considered part of the preclinical phase of major neurocognitive disorder, reported reduced mental HRQoL compared to age-matched controls. It is possible that perception of cognitive difficulties is associated with psychological distress (56), which may in turn lead to decreased mental HRQoL. Our result raises the importance of assessing and promoting subjective cognitive functioning in order to improve HRQoL in older adults with TBI reporting cognitive difficulties. In this study, cognitive functioning was not related to physical HRQoL (effect sizes below the moderate effect threshold – Table 3). In contrast, Gorgoraptis et al. (57)

showed that individuals aged 22–91 years with significant objective cognitive impairments post-TBI reported poorer physical and mental HRQoL than cognitively intact individuals. However, it appears that the association between cognitive functioning and physical HRQoL in their sample was mediated by poor sleep quality, which was not assessed in our study. Finally, results of the present study revealed that cognitive functioning was not related to social participation (effect sizes below the moderate effect threshold – Table 3). This highlights that older adults with TBI may still engage in productive activities, social relationships, and community activities. These results contrast with observations in younger adults with TBI that lower cognitive functioning is associated with lower social participation (28,29). It is possible that cognitive issues may be more readily accepted by older adults, and thus less likely to influence their social activities and community engagement. Older adults are also generally less engaged in full-time work or other productive activities where cognitive issues are more likely to emerge.

The present findings have notable clinical implications. It is essential to consider both subjective perception and objective data in post-TBI cognitive assessment given that despite the fact they are associated with each other they both are related to mental HRQoL. This underscores the clear need to include interventions to improve subjective and objective cognitive functioning during the rehabilitation process and beyond. Some studies conducted with older adults with cognitive complaints but without TBI have suggested that the combination of cognitive training, cognitive restructuring, and psychoeducation is promising in promoting cognitive functioning (58). Interestingly, some authors have suggested that cognitive training may not only have positive outcomes on subjective memory performance but also on perceived psychological well-being (58), which is in line with our finding linking subjective cognitive functioning and mental HRQoL. Future studies should investigate the effectiveness of such interventions at promoting subjective cognitive functioning and HRQoL in older adults with TBI. Additional research is also needed to confirm and expand our findings, for instance by using a more exhaustive neuropsychological assessment or by including a control group to better separate the effects of normal aging and of TBI on cognition. Also, the validity and utility of the many different versions of the TICS-m, ranging from 27 to 50 points (59), should be examined to standardize administration procedures and facilitate comparisons between studies.

The results of the present study must be interpreted in light of some limitations. First, no control group was included and the sample size was somewhat limited. Statistical power may have been insufficient, and the examination of effect sizes suggests that only moderate effects ($r \geq 0.30$) reached statistical significance (see Tables 2 and 3). Second, objective cognitive functioning was assessed with short telephone tests that did not capture all domains of cognition. Third, the F-TICS-m performance might have been slightly underestimated as the number of presentations of the word list differed from the validation study to which results were compared. Moreover, comparisons were made between our study and validation normative values with distinct characteristics samples for the TICS (older

women) and the MOS-Cog (18–98 years old). Fourth, findings cannot be generalized to the whole TBI population as both ends of the injury severity spectrum may have been underrepresented in our study (only hospitalized patients were enrolled, and only one participant had a severe TBI). Also, we cannot exclude the possibility that individuals who were missed or refused to participate had different cognitive outcomes than those recruited. Despite these limitations, this study has several strengths, notably the focus on the elderly population, a systematic recruitment procedure at a level 1 trauma center, the inclusion of objective and subjective cognitive measures, and the exploration of the relationship with more global outcomes such as HRQoL and social participation. In conclusion, this research has provided a better understanding of subjectively and objectively measured cognitive functioning in older adults with TBI. Moreover, it has highlighted the association between cognition and mental HRQoL, a relationship that could be explored further to enhance the wellbeing of this growing population.

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